Abiodun M. Musbaudeen

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Education

Ph.D. in Economics, Binghamton University (SUNY) — Expected May 2027
Dissertation: *Essays on Digitalization, Productivity, and Markup*
Committee: Subal C. Kumbhakar (chair), Ivan Korolev, Yingjun Su
M.Sc. in Economics, University of Lagos, Nigeria — 2017
B.Sc. in Economics, University of Lagos, Nigeria — 2014

Research Fields

Applied Microeconomics; Production Econometrics; Industrial Organization; Applied Econometrics.

Teaching Fields

Principles and intermediate microeconomics and macroeconomics; statistics for economists; introductory and intermediate econometrics.

Publications

Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects
Ivan Korolev, Abiodun Musbaudeen, and Nency Dhameja. 2026.
Economics Letters 268: 113205. doi:10.1016/j.econlet.2026.113205

The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking
Abiodun M. Musbaudeen and Subal C. Kumbhakar. 2026.
Economic Modelling, 107859. Journal article; replication code.

Job Market Paper

Digitalization, Factor-Biased Productivity, and Market Power
With Subal C. Kumbhakar. Under review.
Develops a method to sequentially estimate labor-augmenting productivity, Hicks-neutral productivity, and markups, allowing for persistence and endogenous digitalization. Extends proxy-variable identification to firms with market power without imposing a particular model of competition. Applies the method to Spanish manufacturing firms (ESEE), 2000–2019, using a multidimensional digitalization index. Digitalization is associated with gains in both productivity components, with larger labor-augmenting responses for small firms and smaller markup responses in highly digitalized industries.

Working Papers

Mediation Analysis in Production Functions with Unobserved Productivity
With Subal C. Kumbhakar.

Work in Progress

Regularization in Translog Production Models
Solo-authored.

Artificial Intelligence and Spatial Factor-Biased Productivity
Solo-authored.

Honors and Awards

Best Third-Year Paper Award (co-recipient), Department of Economics, Binghamton University — 2025
Graduate Teaching and Research Assistantship, Binghamton University — 2022–2027

Teaching Experience

Department of Economics, Binghamton University (SUNY)

Instructor

Statistical Methods — Summer 2025

Teaching Assistant

Introduction to Econometrics — Fall 2026; Fall 2025
Lead econometrics TA sessions and R instruction; hold office hours.
Advanced Data Analytics — Spring 2026
Principles of Macroeconomics — Spring 2025
Corporate Economics — Fall 2024
Principles of Microeconomics — Spring 2024
Introduction to Microeconomics — Fall 2023
Principles of Macroeconomics — Spring 2023
Intermediate Microeconomics — Fall 2022

Conferences and Seminars

Southern Economic Association, 96th Annual Meeting *(scheduled)* — November 21–23, 2026
Houston, Texas.
North American Productivity Workshop XII (NAPW) — June 9–12, 2025
Virginia Tech Research Center, Arlington, Virginia.

Professional Experience

Data Engineer and Business Intelligence Analyst, Mindshare (WPP Media), New York — 2021–2022

- Built 43 ETL pipelines consolidating international campaign and spending data; unified reporting across social-media and Adobe Analytics sources and automated Tableau and Power BI reports for global decision-makers.

Head of Data Science and Engineering, Konga eCommerce Group, Nigeria — 2021

- Led a five-person data science and engineering team; directed machine-learning customer segmentation and translated results into targeting and engagement recommendations for senior management.

Senior Data Scientist and Data Engineer, First Bank of Nigeria Ltd., Nigeria — 2018–2021

- Built more than 50 data pipelines and applied machine learning to segment approximately 23 million customer accounts; developed dormancy and channel-choice models to inform retention and service decisions.

Data Research Analyst, West Africa Vocational Education, Nigeria — 2018

- Analyzed trainee retention and employment outcomes to inform admissions decisions; designed questionnaires and compared outcomes for program and non-program candidates.

Technical Skills

Programming: R, Python, Stata, SQL, LaTeX, Git.
Data and visualization: Snowflake, Oracle, Azure Data Factory, Azure Synapse Analytics, Tableau, Power BI.
Methods: GMM, instrumental variables, control functions, production and distance functions, stochastic frontier analysis, panel data, causal mediation, causal forests, difference-in-differences, synthetic control, regression discontinuity, bootstrap inference, simulation, and regularization.

References

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